

Project 3. Reinforcement Learning and Inverse Reinforcement Learning

May 7, 2026

Part 1: Reinforcement Learning

Question 1: Reward Function Heat Maps

We visualize the two reward functions as heat maps over the 10×10 state grid. In both reward functions, state 99 (bottom-right corner) is the unique goal state with a positive reward, and all other non-zero states carry negative rewards acting as traps to be avoided.

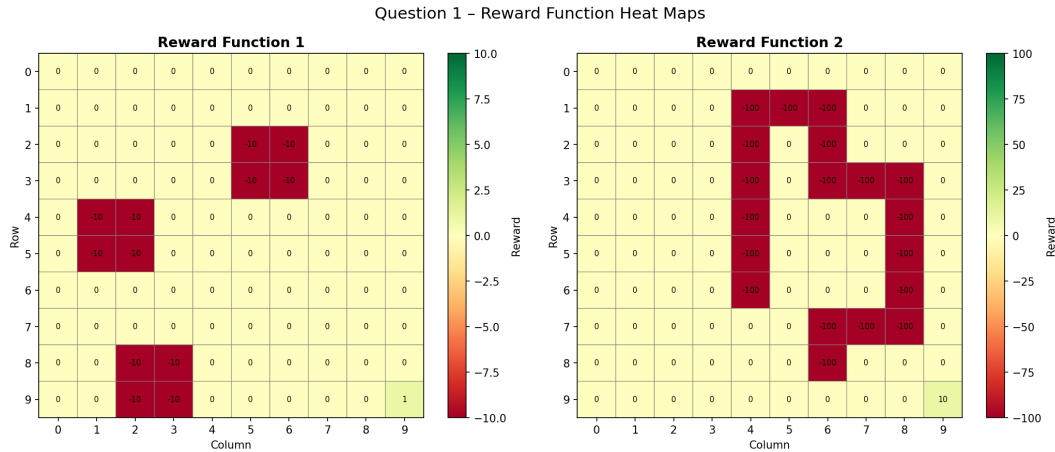


Figure 1: Heat maps of Reward Function 1 (left) and Reward Function 2 (right). Red cells indicate penalty states, green indicates the goal state.

Reward Function 1 contains three clusters of penalty states (each with reward -10) and a goal state with reward $+1$. Reward Function 2 forms a larger U-shaped penalty enclosure (each cell with reward -100) around the center of the grid, with a goal reward of $+10$. Both functions share the same intuition: the agent must navigate to the bottom-right corner while avoiding the negative regions.

Question 2: MDP Setup and Value Iteration

We construct the Markov Decision Process (MDP) with the following parameters:

- Number of states: $|S| = 100$ (10×10 grid)
- Number of actions: $|A| = 4$ (Right, Left, Up, Down)

- Wind probability: $w = 0.1$
- Discount factor: $\gamma = 0.8$
- Reward function: Reward Function 1
- Convergence threshold: $\varepsilon = 0.01$

The transition probabilities follow the rule that each action moves the agent in the intended direction with probability $1 - w$, and moves in a uniformly random direction with probability w . Boundary states remain in place when an action would move the agent off the grid.

We implement the Value Iteration algorithm to find the optimal state-value function $V^*(s)$. The Bellman update at each iteration is:

$$V_{\text{new}}(s) = \max_{a \in A} \sum_{s' \in S} P_{ss'}^a [R(s') + \gamma V(s')].$$

The algorithm is stopped when

$$\max_s |V_{\text{new}}(s) - V(s)| < \varepsilon.$$

The optimal state values upon convergence are shown below.

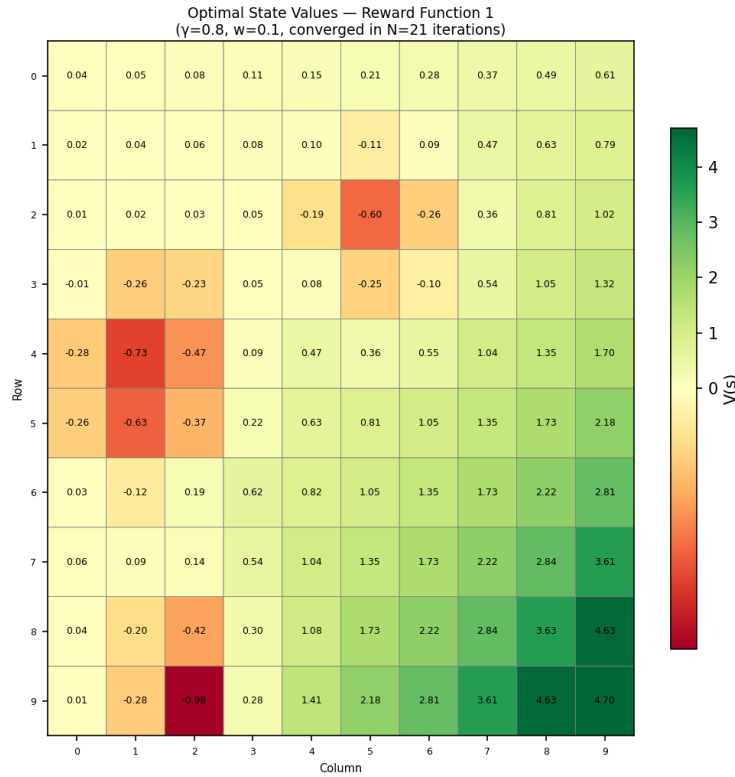


Figure 2: Optimal state values $V^*(s)$ for Reward Function 1 ($\gamma = 0.8$, $w = 0.1$).

Using the convergence-counting convention in which the final convergence-check update is not included as an additional snapshot iteration, the value iteration algorithm converged in $N = 21$ steps. The five snapshot steps are linearly spaced from 1 to N , giving

1, 6, 11, 16, 21.

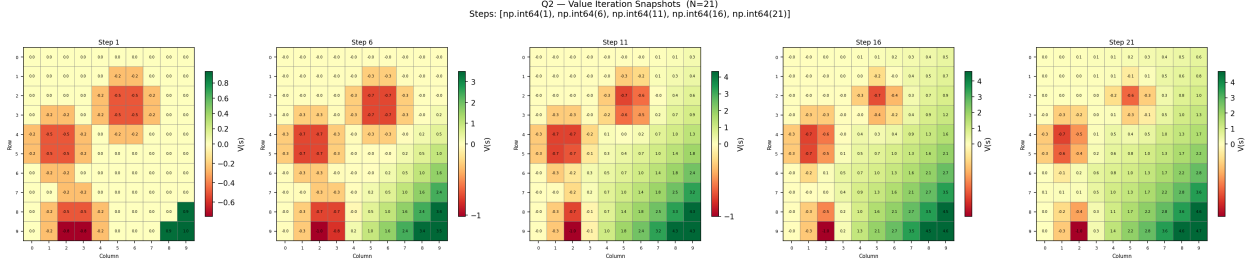


Figure 3: Snapshots of state values at 5 linearly spaced iterations (1, 6, 11, 16, 21). Values propagate outward from the goal and penalty regions as the number of iterations increases.

Observations.

- **Early iterations:** Most state values remain near zero because the reward signal has not yet propagated beyond the immediate neighbors of the goal and penalty states.
- **Mid iterations:** The positive reward from state 99 and the negative rewards from penalty clusters begin to spread outward, creating a visible gradient across the grid.
- **Late iterations:** The value function stabilizes. States near the goal converge to high positive values, while states adjacent to penalty clusters converge to negative values. The overall map becomes smooth as the influence of rewards propagates to all states.

Question 3: Optimal State Value Heat Map

We generate a heat map of the optimal state values $V^*(s)$ obtained from Question 2. The color scale ranges from red (low values) to green (high values), with the colorbar displayed on the right.

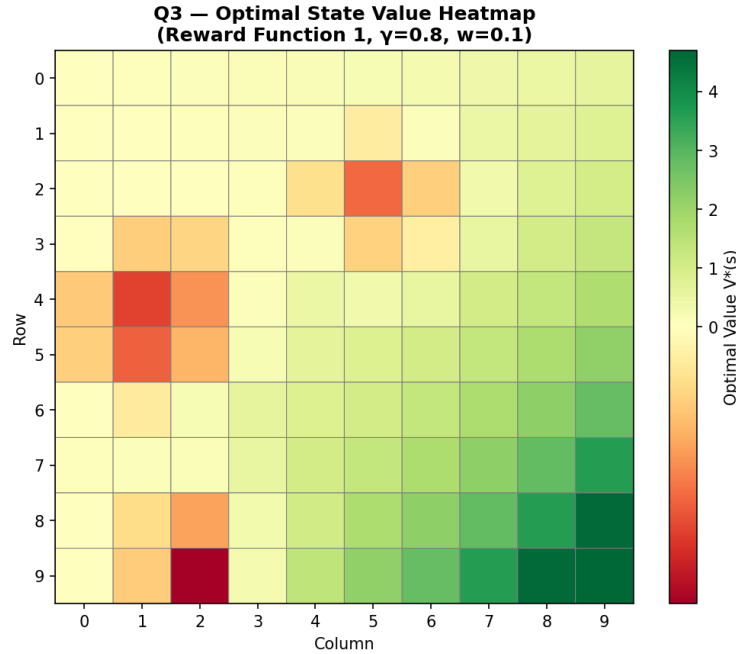


Figure 4: Heat map of optimal state values $V^*(s)$ for Reward Function 1 ($\gamma = 0.8$, $w = 0.1$).

Question 4: Distribution of Optimal State Values

Using the heat map from Question 3, we make the following observations about the distribution of optimal state values across the 2-D grid.

1. Values decrease with distance from the goal. State 99 (bottom-right corner) is the only state with a positive reward (+1), and it achieves the highest optimal value in the grid. States in its vicinity — particularly those in the lower-right quadrant — also carry high values because they can reach the goal in fewer steps. Since the discount factor is $\gamma = 0.8$, each additional step to the goal multiplies the expected return by 0.8, creating a smooth gradient from high values in the bottom-right to lower values in the top-left.

2. Penalty clusters create local value sinks. The three penalty clusters (reward -10 each) are clearly visible as red regions in the heat map:

- Rows 2–3, Columns 5–6
- Rows 4–5, Columns 1–2
- Rows 8–9, Columns 2–3

States adjacent to these clusters also suffer reduced values because the wind probability $w = 0.1$ creates a non-zero chance of being blown into a penalty cell even when the agent attempts to avoid it.

3. Penalty clusters act as barriers. States that lie on the most direct path to the goal but are blocked by a penalty cluster are forced to take longer detours, accumulating more discounting steps and thus receiving lower optimal values. This is especially visible for states in the lower-left region, which are hemmed in by the penalty cluster at rows 8–9 and must navigate around it to reach state 99.

Summary. The heat map shows a value gradient flowing from the bottom-right (highest) toward the top-left (lowest), interrupted by three red sinks at the penalty clusters. The optimal agent is therefore steered toward the right side of the grid, bypassing the penalty regions while converging toward state 99.

Question 5: Optimal Policy (Reward Function 1)

We implement the computation step of the Value Iteration algorithm (lines 14–17 of the pseudocode) to derive the optimal policy:

$$\pi(s) = \arg \max_{a \in A} \sum_{s' \in S} P_{ss'}^a [R(s') + \gamma V^*(s')]$$

The resulting policy is visualized below, where each cell displays an arrow indicating the optimal action at that state.

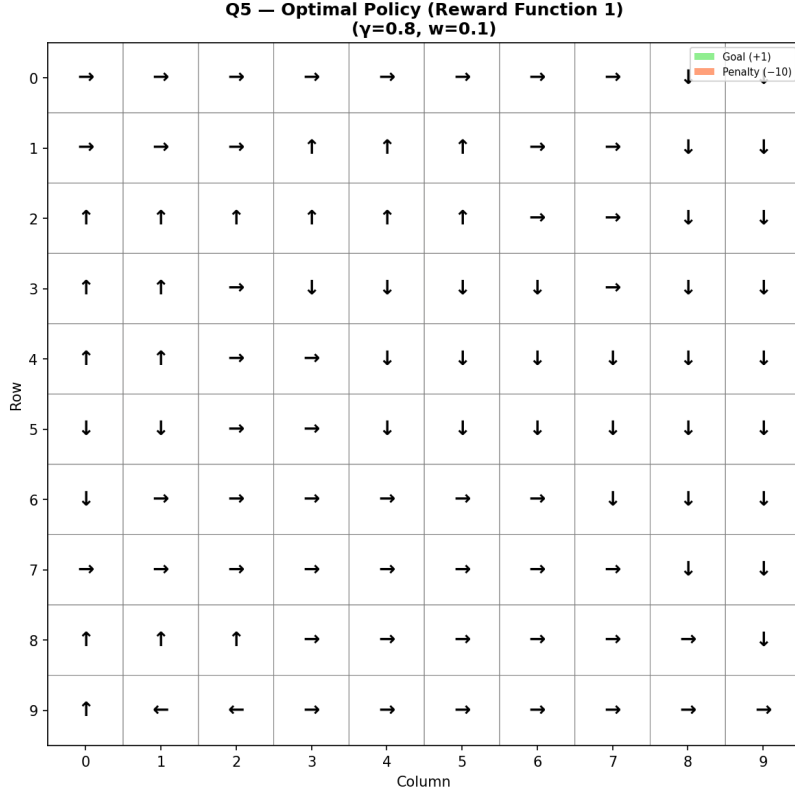


Figure 5: Optimal policy for Reward Function 1 ($\gamma = 0.8, w = 0.1$). Arrows indicate the optimal action at each state.

Does the policy match intuition? Yes. The dominant directions across the grid are right ($\rightarrow$) and down ($\downarrow$), steering the agent toward the goal at state 99 (bottom-right corner). States near the penalty clusters point away from them rather than through them, even when a shorter direct path exists, because the expected penalty outweighs the cost of a longer detour. States near the grid boundary occasionally show unexpected directions due to the wind probability $w = 0.1$, which creates a small but non-zero chance of being blown into a penalty cell. Overall, the policy correctly reflects the trade-off between reaching the goal and avoiding the negative reward regions.

Can the agent determine the optimal action from neighboring values? Yes. Given the known transition probabilities $P_{ss'}^a$ and reward function R , the agent can compute the action-value $Q(s, a)$ using only the optimal values of its neighboring states:

$$Q(s, a) = \sum_{s'} P_{ss'}^a [R(s') + \gamma V^*(s')]$$

Since each action only leads to neighboring states (with small wind perturbations), $V^*(s')$ is needed only for the immediate neighbors of s . Taking $\arg \max_a Q(s, a)$ then gives the optimal action directly, without re-running Value Iteration.

Question 6: Optimal State Values (Reward Function 2)

We replace Reward Function 1 with Reward Function 2 and re-run the same Value Iteration algorithm from Question 2. Using the same convergence-counting convention as in Question 2, the

algorithm converges in $N = 31$ iterations. The optimal state values are shown below.

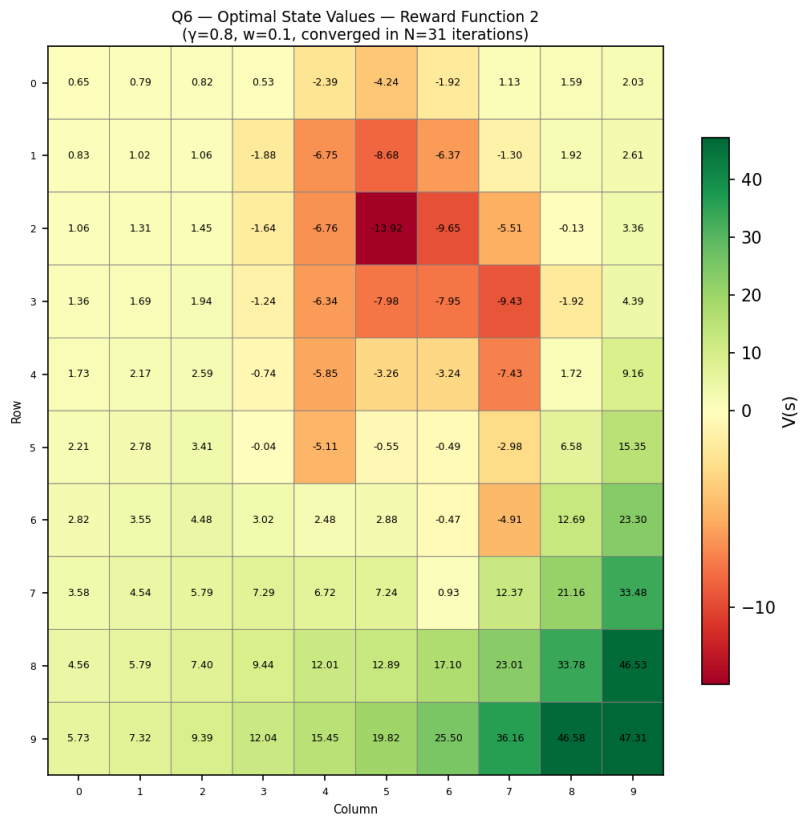


Figure 6: Optimal state values $V^*(s)$ for Reward Function 2 ($\gamma = 0.8$, $w = 0.1$, converged in $N = 31$ iterations).

Question 7: Optimal State Value Heat Map and Distribution (Reward Function 2)

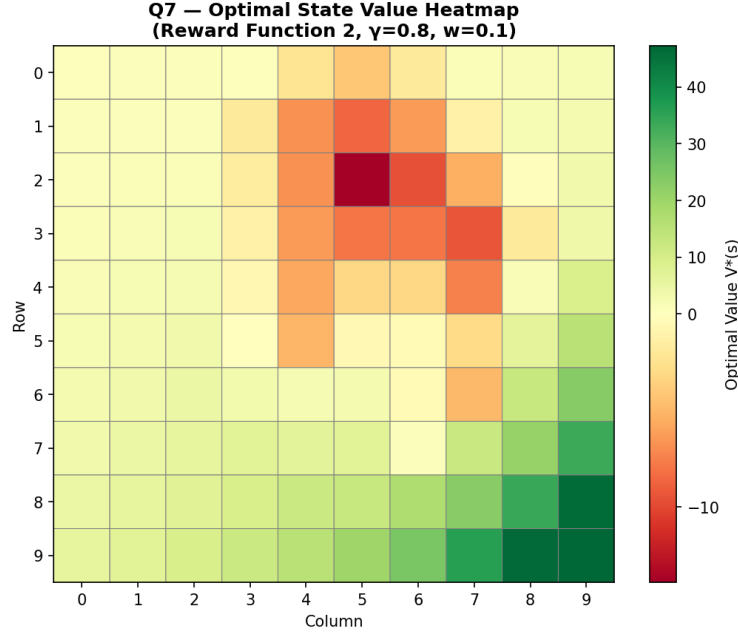


Figure 7: Heat map of optimal state values $V^*(s)$ for Reward Function 2 ($\gamma = 0.8$, $w = 0.1$).

1. Goal state dominates. State 99 (bottom-right corner) has the highest optimal value (+47.31), which is significantly larger than in Reward Function 1 due to the higher goal reward of +10. The surrounding bottom-right region also carries high values, reflecting its proximity to the goal and the absence of nearby penalties.

2. The penalty enclosure creates a large danger zone. The U-shaped penalty enclosure, with reward -100 per penalty cell, produces a large low-value region spanning roughly columns 4–8 and rows 1–8. State 52 (row 2, column 5) has the lowest value (-13.92) among all states because it is nearly surrounded by penalty cells, making it highly likely that the agent will incur a large negative reward regardless of which action it takes.

3. Two distinct corridors emerge. The heat map separates the grid into two relatively safe regions:

- **Left corridor** (columns 0–3): States in this region have moderate positive values and can reach the goal by moving downward first and then turning right around the penalty enclosure.
- **Right corridor** (column 9): States in the rightmost column have a clear path downward toward the goal, yielding steadily increasing values from top to bottom.

4. Wind widens the danger zone. With wind probability $w = 0.1$, states one step outside the penalty boundary also receive reduced values, since there is a non-zero probability of being blown into a -100 penalty cell. This causes the low-value region to extend slightly beyond the actual penalty cells.

Summary. Compared to Reward Function 1, the value distribution under Reward Function 2 is more extreme: the goal region is much brighter, while the penalty enclosure creates a much darker danger zone. The structured U-shaped penalty region forces the agent into narrow safe corridors on the left and right sides of the grid, which is clearly reflected in the sharp contrast of the heat map.

Question 8: Optimal Policy (Reward Function 2)

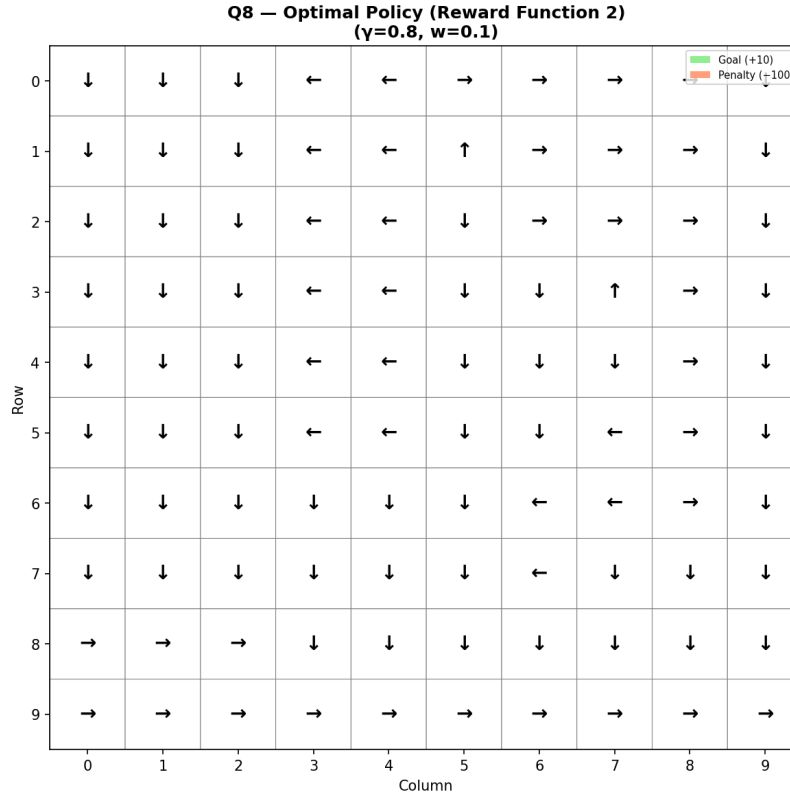


Figure 8: Optimal policy for Reward Function 2 ($\gamma = 0.8, w = 0.1$). Arrows indicate the optimal action at each state.

Does the policy match intuition? Yes. The policy clearly reflects the structure of the U-shaped penalty enclosure and the location of the goal at state 99. Two dominant navigation strategies are visible:

- **Left corridor** (columns 0–2): States navigate downward (↓) along the left side of the grid, staying well clear of the penalty enclosure, before turning right (→) along row 8–9 to reach the goal.
- **Right corridor** (column 9): States in the rightmost column navigate straight down (↓) toward the goal, taking the most direct penalty-free path.

States adjacent to the penalty enclosure point sharply away from it, confirming that the -100 penalty magnitude dominates the agent’s decision even when a more direct route to the goal would pass through the enclosure. No state selects an action that leads directly into a penalty cell, which is consistent with the intuition developed from the heat map in Question 7.

Question 9: Effect of Wind Parameter ($w = 0.1$ vs $w = 0.6$)

We change the wind probability to $w = 0.6$ and recompute the optimal policy for both reward functions. The results are compared side by side below.

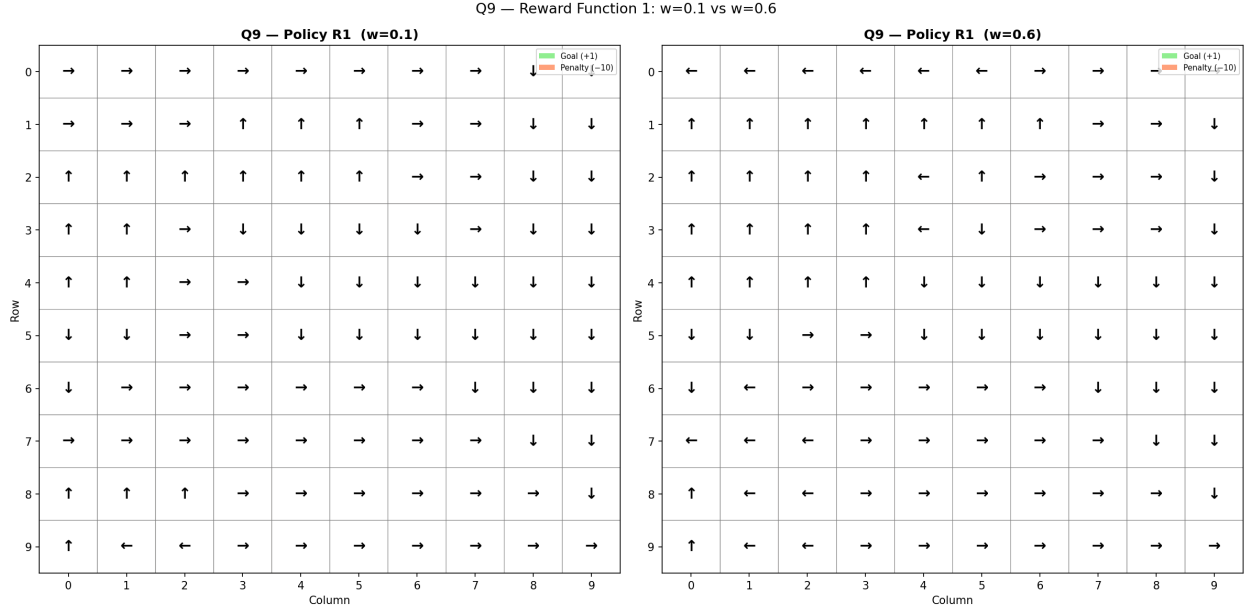


Figure 9: Optimal policy for Reward Function 1 with $w = 0.1$ (left) and $w = 0.6$ (right).

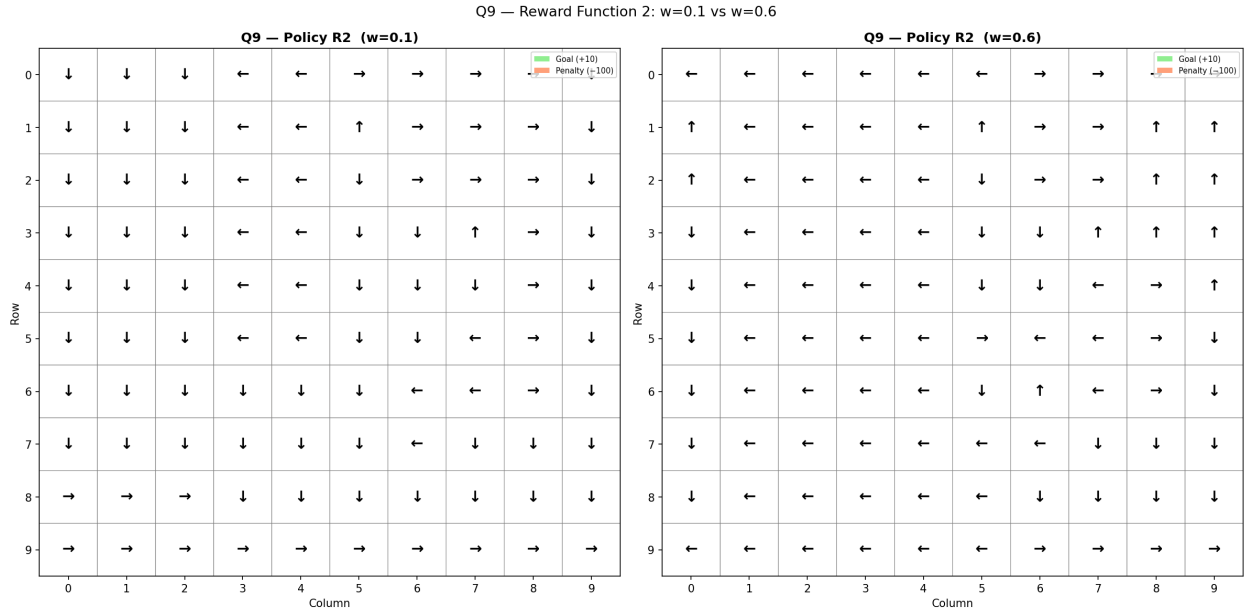


Figure 10: Optimal policy for Reward Function 2 with $w = 0.1$ (left) and $w = 0.6$ (right).

Observed differences. With $w = 0.6$, the agent is blown into a random direction 60% of the time, meaning it has very limited control over its movement. This leads to several notable changes in both policies:

- **Reward Function 1:** Under $w = 0.1$, most states navigate clearly toward the goal via rightward and downward arrows. Under $w = 0.6$, many states in the upper and left regions prefer to move left ($\leftarrow$) or up ($\uparrow$), steering away from penalty clusters rather than toward the goal. The policy becomes more defensive, prioritizing penalty avoidance over goal-seeking.
- **Reward Function 2:** The effect is more pronounced. Under $w = 0.1$, two clear corridors emerge on either side of the penalty enclosure. Under $w = 0.6$, most states point left ($\leftarrow$) or away from the enclosure, with no visible coherent path toward state 99. The high wind makes the risk of landing in a -100 penalty cell so large that the agent prefers to stay far from the enclosure entirely, even at the cost of never reaching the goal.

Choice of w for the IRL section. We select $w = 0.1$ for the remainder of the project. With $w = 0.1$, the optimal policy exhibits a coherent structure that clearly steers the agent toward the goal while avoiding penalties. With $w = 0.6$, the policy becomes overly conservative and loses its navigational structure, making it a poor expert behavior for IRL to recover. A lower wind probability produces a more informative and transferable expert policy.

Part 2: Inverse Reinforcement Learning

Question 10: LP Formulation — Expressing c , x , D , b

Optimization variable x .

$$x = \begin{bmatrix} t \\ u \\ R \end{bmatrix} \in \mathbb{R}^{3|S|}$$

Objective vector c .

$$c = \begin{bmatrix} \mathbf{1}_{|S| \times 1} \\ -\lambda_{|S| \times 1} \\ \mathbf{0}_{|S| \times 1} \end{bmatrix}$$

Constraint matrix D .

$$D = \begin{bmatrix} I_{|S| \times |S|} & \mathbf{0} & (P_a - P_{a_1})(I - \gamma P_{a_1})^{-1} \\ \mathbf{0} & \mathbf{0} & (P_a - P_{a_1})(I - \gamma P_{a_1})^{-1} \\ \mathbf{0} & -I_{|S| \times |S|} & I_{|S| \times |S|} \\ \mathbf{0} & -I_{|S| \times |S|} & -I_{|S| \times |S|} \\ \mathbf{0} & \mathbf{0} & I_{|S| \times |S|} \\ \mathbf{0} & \mathbf{0} & -I_{|S| \times |S|} \end{bmatrix}$$

Constraint vector b .

$$b = \begin{bmatrix} \mathbf{0}_{|S| \times 1} \\ \mathbf{0}_{|S| \times 1} \\ \mathbf{0}_{|S| \times 1} \\ \mathbf{0}_{|S| \times 1} \\ (R_{\max})_{|S| \times 1} \\ (R_{\max})_{|S| \times 1} \end{bmatrix}$$

Question 11: IRL Accuracy vs λ (Reward Function 1)

We sweep λ from 0 to 5 using 500 evenly spaced values. For each λ , we solve the LP in equation (2) to extract the reward function, run Value Iteration on the extracted reward, and compute the accuracy against the expert policy from Question 5.

Although the LP formulation section states that the IRL discount factor is $\gamma = 0.9$, the implementation guide specifies $\gamma = 0.8$. Therefore, we use $\gamma = 0.8$ consistently in the IRL implementation, matching the value used in the Value Iteration experiments.

In the LP implementation, we also apply a row-masking step when constructing the constraints for each comparison action a . Specifically, if the expert policy already selects action a at state s , then the corresponding row is masked out. This avoids the degenerate case where $a = a_1(s)$, for which

$$(P_{a_1}(s) - P_a(s))(I - \gamma P_{a_1})^{-1}R = 0,$$

which would otherwise impose the meaningless constraint $t_s \leq 0$. Therefore, row masking ensures that the margin constraints are only enforced between the expert action and truly different alternative actions.

The accuracy is computed as

$$\text{Accuracy} = \frac{\sum_{s \in S} m(s)}{|S|}, \quad m(s) = \begin{cases} 1, & O_A(s) = O_E(s), \\ 0, & \text{otherwise.} \end{cases}$$

Here, $O_E(s)$ denotes the expert action from the optimal policy in Question 5, and $O_A(s)$ denotes the action obtained by running Value Iteration on the reward function extracted by IRL.

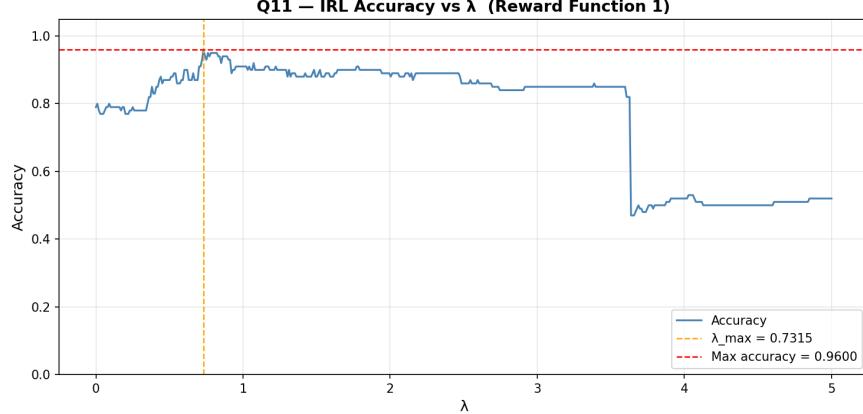


Figure 11: IRL accuracy vs λ for Reward Function 1. The orange dashed line marks $\lambda_{\max}^{(1)} = 0.7315$ and the red dashed line marks the maximum accuracy of 0.96.

The curve exhibits three distinct regions:

- **Low λ ($0 \leq \lambda \lesssim 0.7$):** Accuracy rises as λ increases. A small penalty on reward magnitude acts as regularization, discouraging degenerate reward functions and improving the quality of the extracted policy.
- **Moderate λ ($0.7 \lesssim \lambda \lesssim 3.5$):** Accuracy plateaus around 0.85–0.95. The regularization is strong enough to produce stable reward functions, but the penalty begins to compress the reward magnitudes.

- **High λ ($\lambda \gtrsim 3.5$):** Accuracy drops sharply to around 0.5. Excessive penalization forces $R \rightarrow 0$, collapsing all state values to near zero and making the policy essentially arbitrary.

Question 12: Optimal λ for Reward Function 1

From the plot in Question 11, the value of λ that maximises the IRL accuracy is:

$$\lambda_{\max}^{(1)} = 0.7315$$

At this value, the IRL algorithm achieves a maximum accuracy of 0.96, meaning 96 out of 100 states have the same optimal action as the expert policy from Question 5.

Question 13: Ground Truth vs Extracted Reward Heat Maps (Reward Function 1, $\lambda = \lambda_{\max}^{(1)}$)

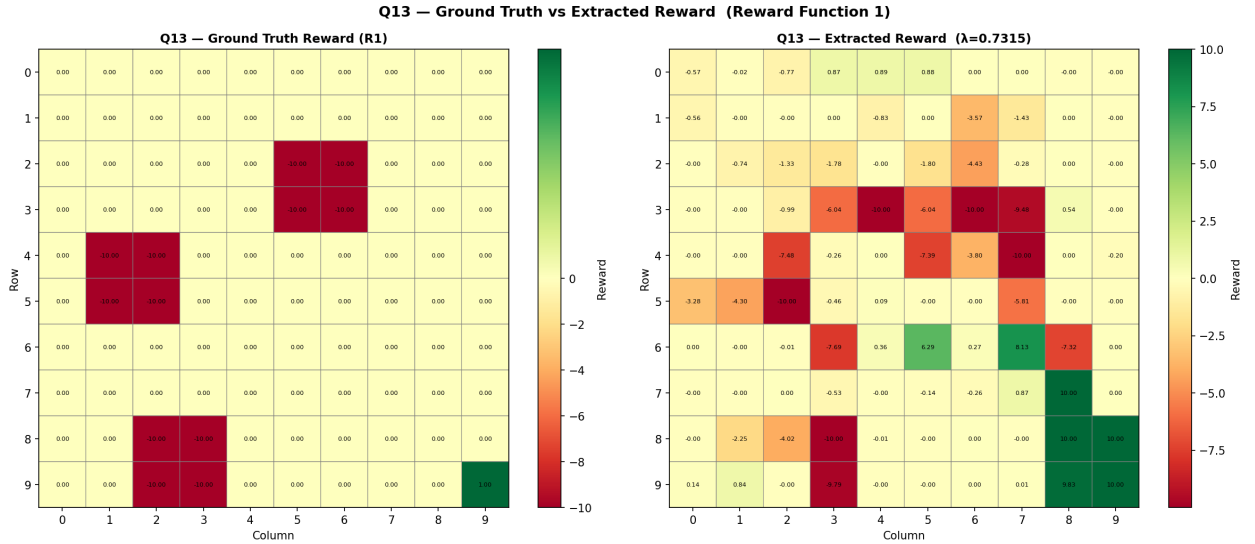


Figure 12: Heat maps of the ground truth reward (left) and the extracted reward at $\lambda_{\max}^{(1)} = 0.7315$ (right) for Reward Function 1.

The ground truth reward is sparse: only 12 penalty states (reward -10) and one goal state (reward $+1$) are non-zero, with all other states receiving zero reward.

The extracted reward is denser and noisier. Several observations can be made:

- **Penalty regions are partially recovered.** The three penalty clusters are visible as red regions in the extracted map, and the magnitudes of several penalty cells are correctly identified at or near -10 .
- **Non-penalty states receive non-zero rewards.** Unlike the ground truth, the extracted reward assigns small positive and negative values to many states that should be zero. This is a consequence of the LP optimizing for policy consistency rather than reward sparsity.
- **High-value states appear near the goal.** Several states in the bottom-right region of the extracted map receive positive rewards (up to $+10$), reflecting the influence of the goal at state 99, even though the ground truth assigns $+1$ only to state 99 itself.

- **Scale differences.** The extracted reward uses the full range $[-R_{\max}, +R_{\max}] = [-10, +10]$, whereas the ground truth has a much narrower effective range. This scale ambiguity is an inherent limitation of the LP-based IRL formulation.

Question 14: Optimal State Values from Extracted Reward ($\lambda = \lambda_{\max}^{(1)}$)

We apply the Value Iteration algorithm from Question 2 to the extracted reward function obtained in Question 13 and generate the corresponding heat map.

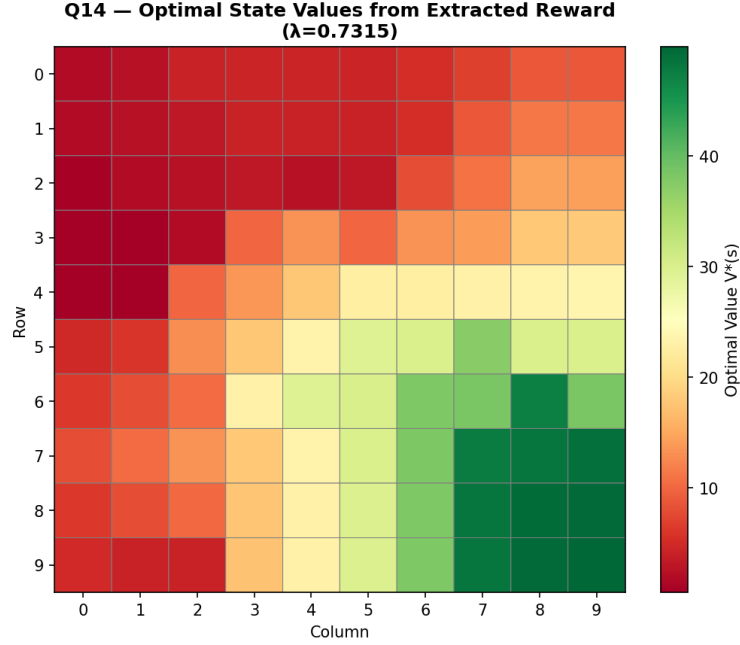


Figure 13: Heat map of optimal state values computed from the extracted reward function at $\lambda_{\max}^{(1)} = 0.7315$.

Question 15: Comparison of Q3 and Q14 Heat Maps

Comparing the optimal state value heat maps from Question 3 (ground truth reward) and Question 14 (extracted reward), we observe the following similarities and differences.

Similarities.

- **Goal region dominates in both maps.** States in the bottom-right corner consistently carry the highest optimal values in both heat maps, reflecting the shared goal at state 99.
- **Values decrease with distance from the goal.** Both maps show a general gradient from high values in the lower-right to lower values in the upper-left, consistent with the discounting effect of $\gamma = 0.8$.

Differences.

- **Scale.** The value range in Q14 is much larger (up to ~ 50) compared to Q3 (up to ~ 5). This is due to the extracted reward assigning larger positive values to states near the goal, amplifying the cumulative discounted return.

- **Penalty sinks are less distinct.** In Q3, the three penalty clusters produce clear local minima visible as dark red patches. In Q14, the extracted reward introduces noise across many states, smoothing out the sharp boundaries between penalty and non-penalty regions.
- **Spatial structure is less defined.** The ground truth heat map in Q3 has well-separated high-value and low-value regions. The extracted heat map in Q14 shows a more gradual and less structured transition, reflecting the imperfect reward recovery by the IRL algorithm.

Question 16: Optimal Policy from Extracted Reward (Reward Function 1, $\lambda = \lambda_{\max}^{(1)}$)

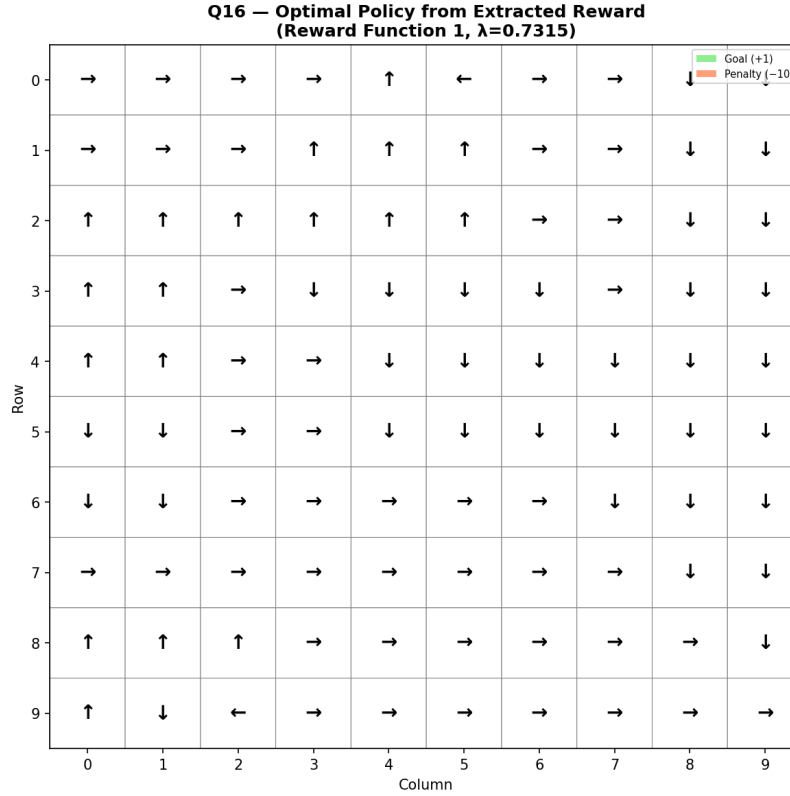
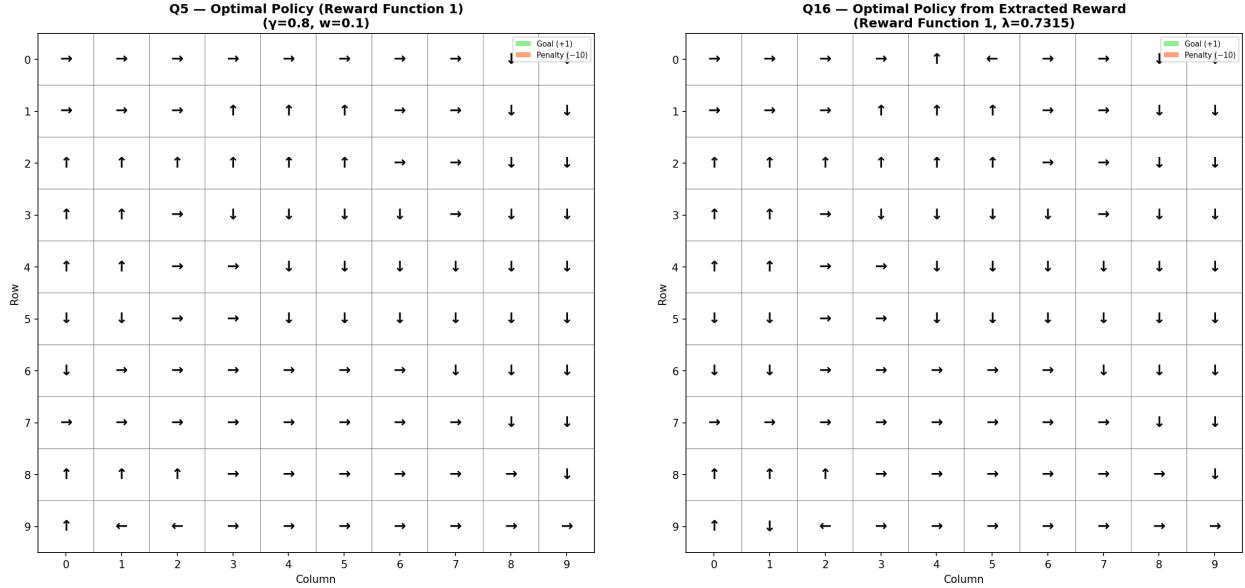


Figure 14: Optimal policy derived from the extracted reward function at $\lambda_{\max}^{(1)} = 0.7315$ for Reward Function 1.

Question 17: Comparison of Q5 and Q16 Policies



(a) Q5: Optimal policy from ground truth reward.

(b) Q16: Optimal policy from extracted reward.

Figure 15: Comparison of optimal policies from Q5 and Q16 (Reward Function 1).

The two policies achieve an accuracy of 0.96, meaning 96 out of 100 states select the same optimal action.

Similarities.

- **Dominant directions are preserved.** Both policies navigate predominantly rightward (→) and downward (↓) toward the goal at state 99. The overall flow structure is nearly identical across the grid.
- **Penalty avoidance is captured.** In both maps, no state selects an action that leads directly into a penalty cell. The extracted reward successfully encodes the need to avoid the three penalty clusters.

Differences.

- **A few states disagree near penalty boundaries.** The 4 mismatched states are concentrated near the edges of the penalty clusters, where the Q-values of competing actions are close. The noisy extracted reward shifts the balance slightly, causing a different action to be selected.
- **Some boundary states show off-grid actions.** A small number of states in Q16 point toward the grid boundary (e.g. row 0 col 5 points ↑, row 9 col 1 points ↓), which keeps the agent in place. These do not appear in Q5 and are caused by the extracted reward assigning slightly higher value to the current state than its neighbors in those locations.

Question 18: IRL Accuracy vs λ (Reward Function 2)

We repeat the same λ sweep from Question 11 using the expert policy from Question 8 (Reward Function 2) as $O_E(s)$.

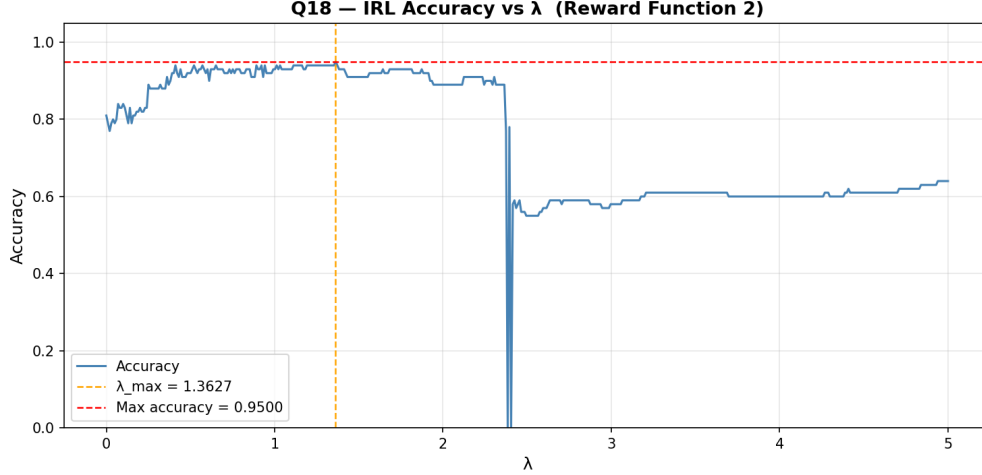


Figure 16: IRL accuracy vs λ for Reward Function 2. The orange dashed line marks $\lambda_{\max}^{(2)} = 1.3627$ and the red dashed line marks the maximum accuracy of 0.95.

The curve shows a similar three-region structure to Question 11:

- **Low λ** ($0 \leq \lambda \lesssim 1.4$): Accuracy rises steadily from around 0.80 to 0.95 as regularization stabilises the extracted reward function.
- **Moderate λ** ($1.4 \lesssim \lambda \lesssim 2.5$): Accuracy plateaus around 0.90–0.95 before dropping sharply near $\lambda \approx 2.5$.
- **High λ** ($\lambda \gtrsim 2.5$): Accuracy collapses to around 0.55–0.65 as excessive penalization forces $R \rightarrow 0$.

Compared to Reward Function 1, the peak accuracy is slightly lower (0.95 vs 0.96) and the optimal λ is higher (1.3627 vs 0.7315), reflecting the greater complexity of recovering the U-shaped penalty enclosure from behavioral observations alone.

Question 19: Optimal λ for Reward Function 2

From the plot in Question 18, the value of λ that maximises the IRL accuracy is:

$$\lambda_{\max}^{(2)} = 1.3627$$

At this value, the IRL algorithm achieves a maximum accuracy of 0.95, meaning 95 out of 100 states have the same optimal action as the expert policy from Question 8.

Question 20: Ground Truth vs Extracted Reward Heat Maps (Reward Function 2, $\lambda = \lambda_{\max}^{(2)}$)

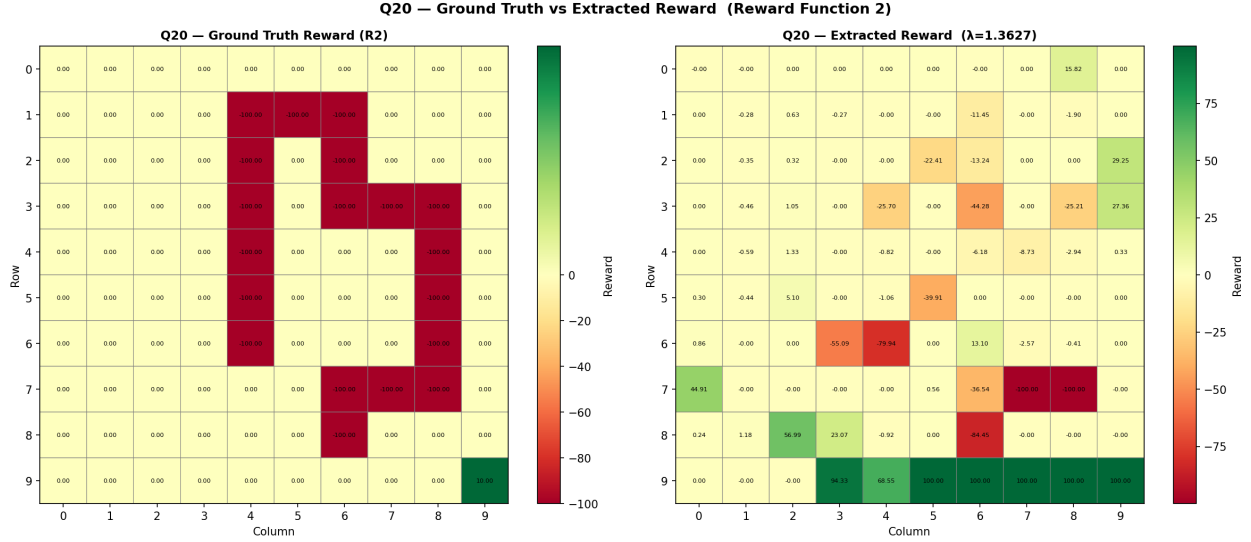


Figure 17: Heat maps of the ground truth reward (left) and the extracted reward at $\lambda_{\max}^{(2)} = 1.3627$ (right) for Reward Function 2.

The ground truth reward is sparse, with -100 penalties forming a U-shaped enclosure and a single goal reward of $+10$ at state 99.

The extracted reward is substantially noisier and differs from the ground truth in several ways:

- **Penalty structure is partially recovered.** Several cells within the U-shaped enclosure receive large negative values (e.g. -79.94 , -84.45 , -100.00), indicating that the IRL algorithm correctly identifies the penalty region. However, not all penalty cells are recovered, and the spatial boundary of the enclosure is less clearly defined than in the ground truth.
- **Large positive rewards appear in the bottom rows.** Row 9 receives very large positive extracted rewards (up to $+100$) in columns 3–9, which do not exist in the ground truth. This occurs because the LP assigns high rewards to states that the expert policy frequently visits on the way to the goal, concentrating positive values near the bottom of the grid.
- **Non-penalty states receive non-zero rewards.** Many states that should have zero reward in the ground truth carry small positive or negative values in the extracted map, introducing noise that slightly distorts the resulting optimal policy.
- **Scale is fully utilized.** The extracted reward spans the full range $[-R_{\max}, +R_{\max}] = [-100, +100]$, whereas the ground truth uses only $\{-100, 0, +10\}$. This scale ambiguity is an inherent limitation of the LP-based IRL formulation.

Question 21: Optimal State Values from Extracted Reward (Reward Function 2, $\lambda = \lambda_{\max}^{(2)}$)

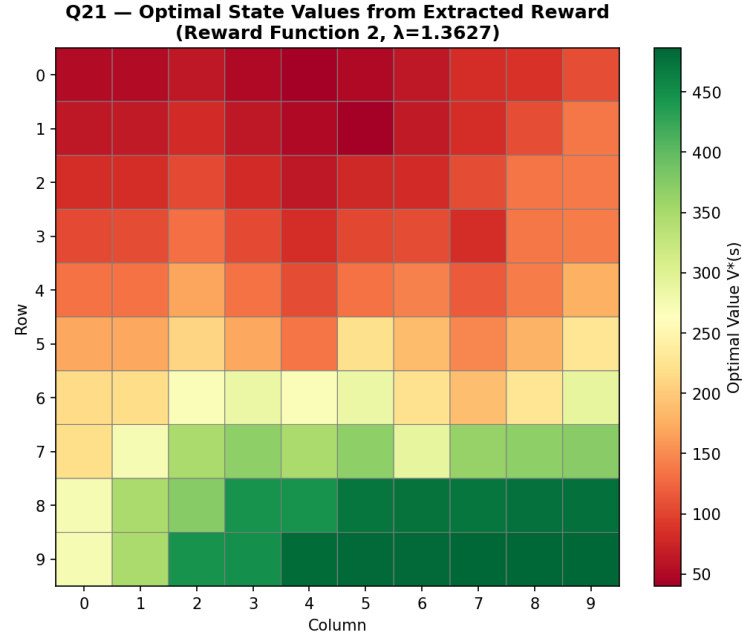
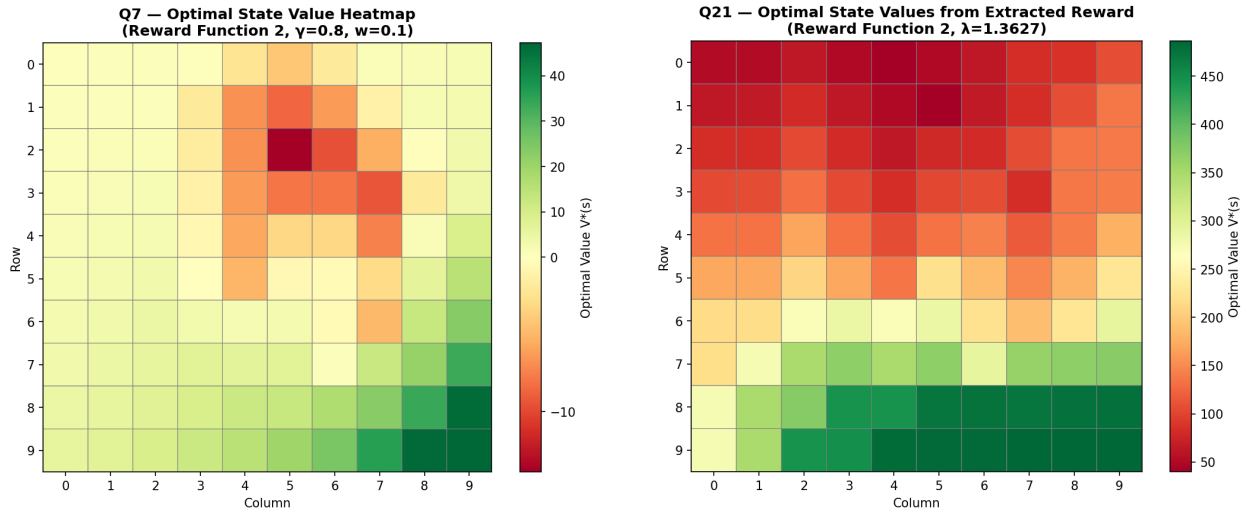


Figure 18: Heat map of optimal state values computed from the extracted reward function at $\lambda_{\max}^{(2)} = 1.3627$ for Reward Function 2.

Question 22: Comparison of Q7 and Q21 Heat Maps



(a) Q7: Optimal values from ground truth reward. (b) Q21: Optimal values from extracted reward.

Figure 19: Comparison of optimal state value heat maps from Q7 and Q21 (Reward Function 2).

Similarities.

- **High-value regions appear near the bottom of the grid.** In both heat maps, the lower part of the grid carries relatively high optimal values. In Q7, this is mainly because these states are close to the true goal at state 99. In Q21, this effect becomes even stronger because the extracted reward assigns large positive rewards to several bottom-row states.
- **Both maps still favor movement toward the goal region.** Although the extracted reward is noisier than the ground truth reward, the resulting value function still assigns larger values to states that are closer to the bottom-right goal region.

Differences.

- **Scale differs significantly.** The Q7 map ranges from approximately -14 to $+47$, whereas the Q21 map ranges from roughly $+50$ to $+475$. This large upward shift is caused by the extracted reward assigning large positive values to bottom-row states, which then propagate high cumulative discounted returns throughout the grid.
- **The penalty enclosure is much less distinct in Q21.** In Q7, the U-shaped penalty enclosure creates a clearly visible low-value region in the center of the grid. In Q21, this structure is less apparent because the extracted reward does not perfectly recover all penalty cells, and the large positive rewards near the bottom of the grid dominate the value function.
- **The value gradient direction differs.** In Q7, values generally increase toward the bottom-right corner, while being strongly distorted by the U-shaped penalty enclosure. In Q21, the gradient runs more strongly from top to bottom, because the extracted reward places high positive rewards across the lower rows rather than only at the true goal state.
- **Q21 is less faithful to the ground-truth spatial structure.** The ground-truth value map in Q7 clearly reflects both the true goal location and the U-shaped penalty region. By contrast, the extracted-reward value map in Q21 is dominated by the reward artifacts produced by IRL, especially the large positive rewards assigned near the bottom of the grid.

Question 23: Optimal Policy from Extracted Reward (Reward Function 2, $\lambda = \lambda_{\max}^{(2)}$)

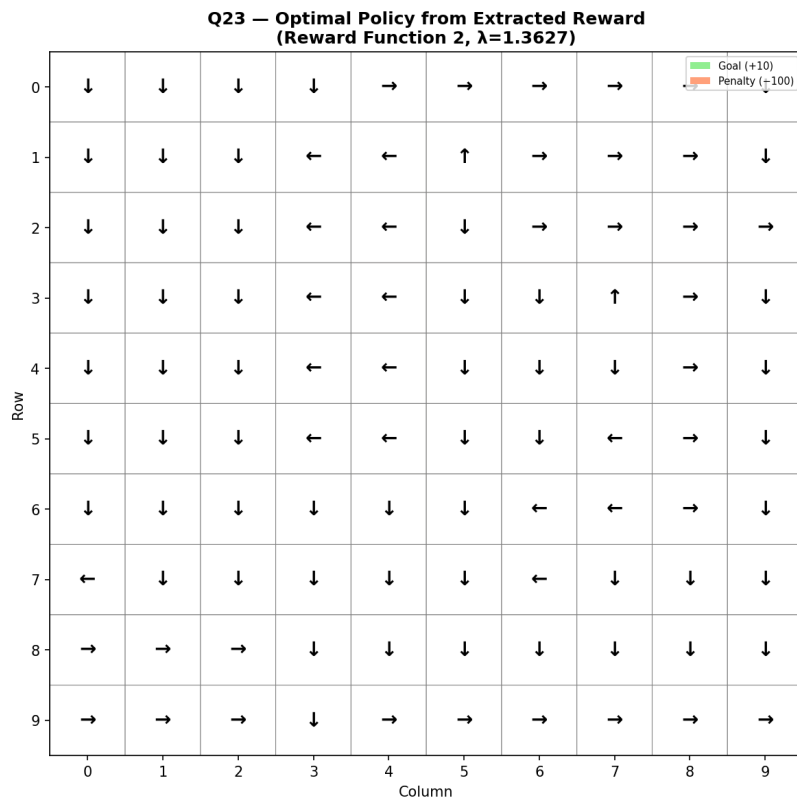
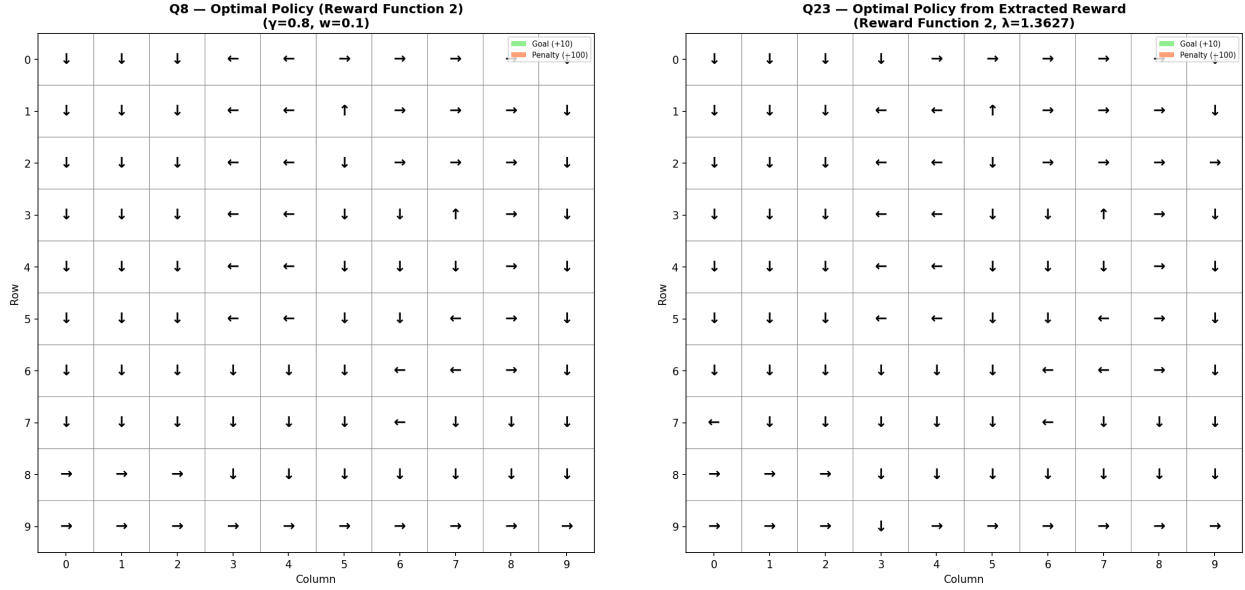


Figure 20: Optimal policy derived from the extracted reward function at $\lambda_{\max}^{(2)} = 1.3627$ for Reward Function 2.

Question 24: Comparison of Q9 and Q23 Policies



(a) Q9: Optimal policy from ground truth reward.

(b) Q23: Optimal policy from extracted reward.

Figure 21: Comparison of optimal policies from Q9 and Q23 (Reward Function 2).

The two policies achieve an accuracy of 0.95, meaning 95 out of 100 states select the same optimal action.

Similarities.

- **Left corridor is preserved.** Both policies navigate the agent downward (↓) along columns 0–2, staying clear of the penalty enclosure before turning right toward state 99.
- **Penalty avoidance is maintained.** States adjacent to the U-shaped penalty enclosure point away from it in both maps, indicating that the extracted reward correctly captures the dangerous regions.
- **Right column navigates toward goal.** Column 9 points consistently downward (↓) in both policies, taking the most direct penalty-free path to the goal.

Differences.

- **Boundary states show stuck actions.** In Q23, a small number of boundary states point into the grid edge, keeping the agent in place. For example, row 7 col 0 points left (←) from the leftmost column, and row 9 col 3 points down (↓) from the bottom row. Neither of these appears in Q8. These errors arise because the goal state is not treated as absorbing in the value iteration, distorting the value of nearby boundary states.
- **A few states near the penalty boundary disagree.** The remaining mismatches are concentrated near the edges of the U-shaped enclosure, where the extracted reward does not precisely recover the -100 penalty magnitudes, causing small Q-value differences to resolve in a different direction.

Question 25: Two Major Discrepancies and Fix

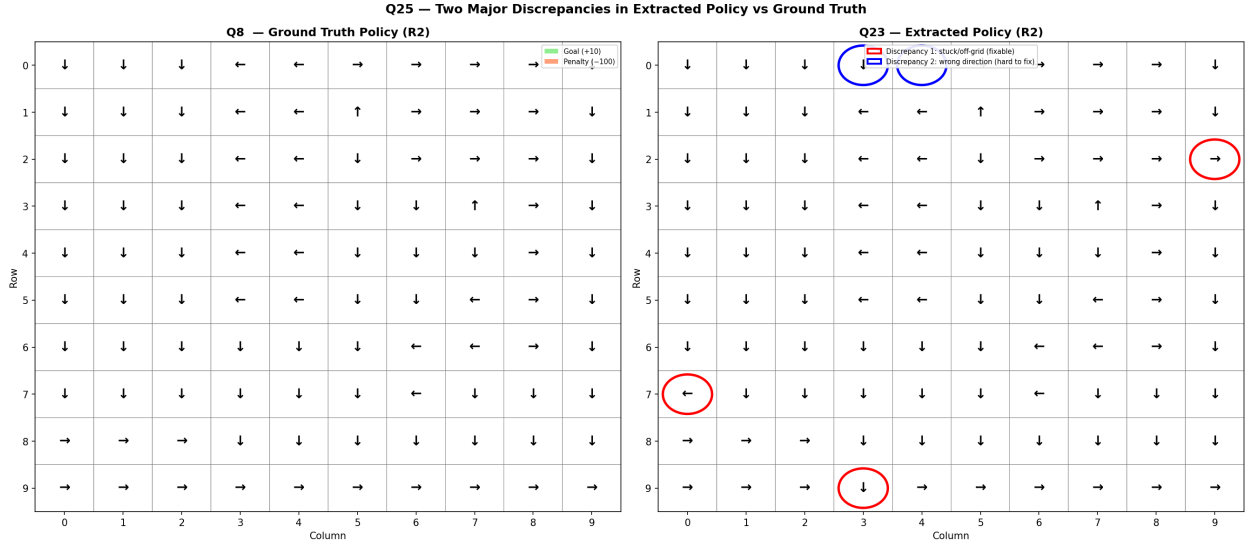


Figure 22: Comparison of Q8 (ground truth, left) and Q23 (extracted, right). Red circles mark Discrepancy 1 (stuck/off-grid actions) and blue circles mark Discrepancy 2 (wrong direction).

Identifying the two discrepancies. Discrepancy 1 (Fixable): Goal state not absorbing. Two states in Q23 point into the grid boundary, keeping the agent stuck indefinitely: row 7 col 0 points $\leftarrow$ (leftmost column) and row 9 col 3 points $\downarrow$ (bottom row). Since both actions hit the boundary, the agent remains in place and never reaches the goal. This occurs because the value iteration does not treat state 99 as an absorbing (terminal) state. Without enforcing the absorbing condition, $V(99)$ is computed using future transitions out of state 99 via wind, distorting the values of nearby boundary states and causing incorrect arg max choices.

Discrepancy 2 (Hard to fix): IRL ill-posedness and reward equivalence. Row 2 col 9 points $\rightarrow$ (off the right boundary) in Q23 instead of $\downarrow$ as in Q8. This error cannot be resolved by modifying the value iteration, as it stems from a fundamental theoretical limitation of IRL itself.

IRL is an *ill-posed* inverse problem: for any extracted reward R , the family of reward functions obtained via potential-based reward shaping,

$$R'(s) = R(s) + f(s) - \gamma \mathbb{E}[f(s')],$$

all induce the same optimal policy as R . This means the LP formulation can only recover a reward function up to such an equivalence class — it cannot pinpoint the true ground-truth reward among the infinitely many rewards that are consistent with the observed expert behavior.

As a consequence, the extracted reward may assign values to boundary states that differ from the ground truth in both magnitude and sign, even while producing the correct policy on the majority of states. In this case, the extracted penalties near the right boundary are insufficient to prevent the agent from preferring the boundary action at row 2 col 9, because the Q-values of $\rightarrow$ and $\downarrow$ differ by only a small amount under the extracted reward. Since the LP cannot recover the exact reward magnitudes, these small Q-value differences are susceptible to tie-breaking errors that do not appear under the true reward function. Resolving this discrepancy would require a more constrained IRL formulation, such as Maximum Entropy IRL or Bayesian IRL, which impose additional structure on the reward function beyond mere policy consistency.

Fix: Enforcing the absorbing goal state. We modify the value iteration to enforce the absorbing condition after every Bellman update:

$$V(\text{goal}) = \frac{R(\text{goal})}{1 - \gamma}$$

This correctly reflects that an absorbing goal state with constant reward $R(\text{goal})$ yields the geometric series $R + \gamma R + \gamma^2 R + \dots = R/(1 - \gamma)$.

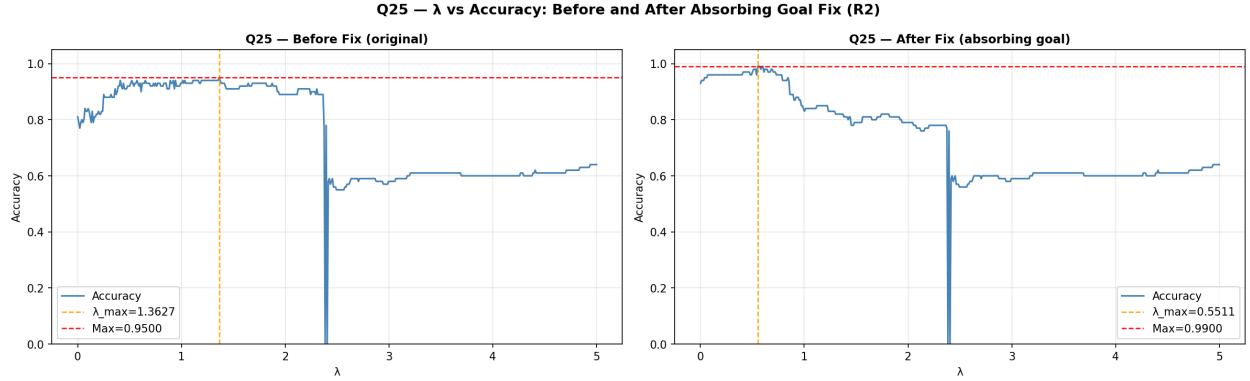


Figure 23: λ vs accuracy before (left) and after (right) the absorbing goal fix. Maximum accuracy improves from 0.95 to 0.99.

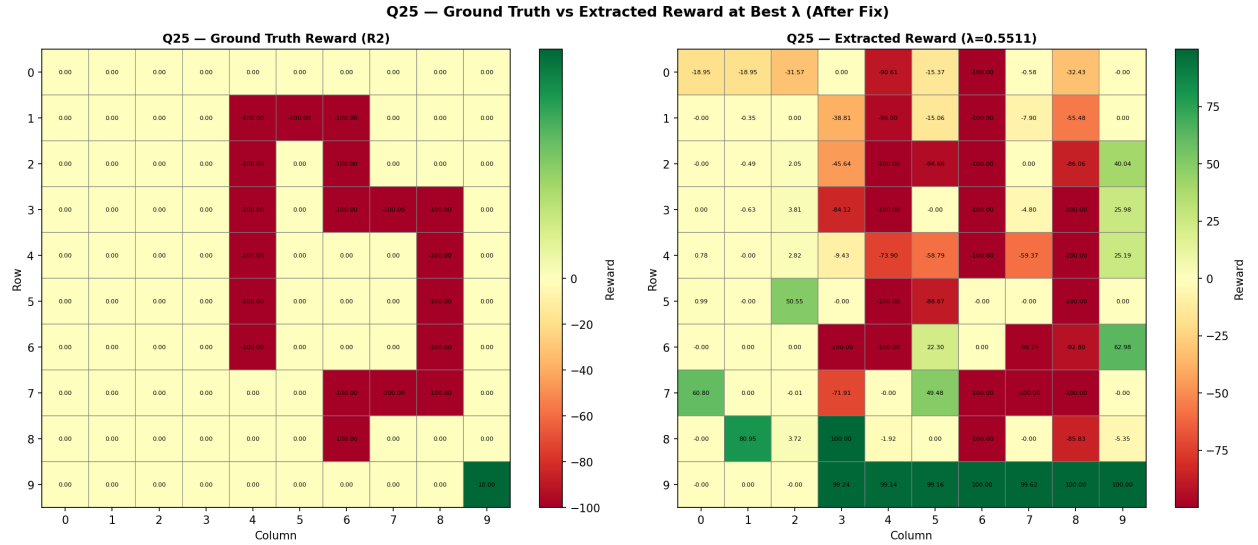


Figure 24: Ground truth reward (left) and extracted reward at $\lambda = 0.5511$ after the fix (right).

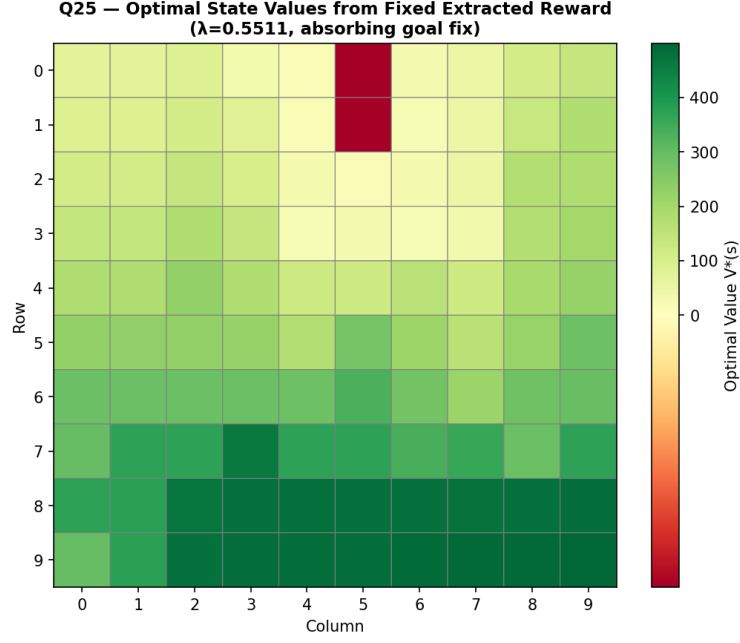


Figure 25: Optimal state values computed from the extracted reward after the absorbing goal fix ($\lambda = 0.5511$).

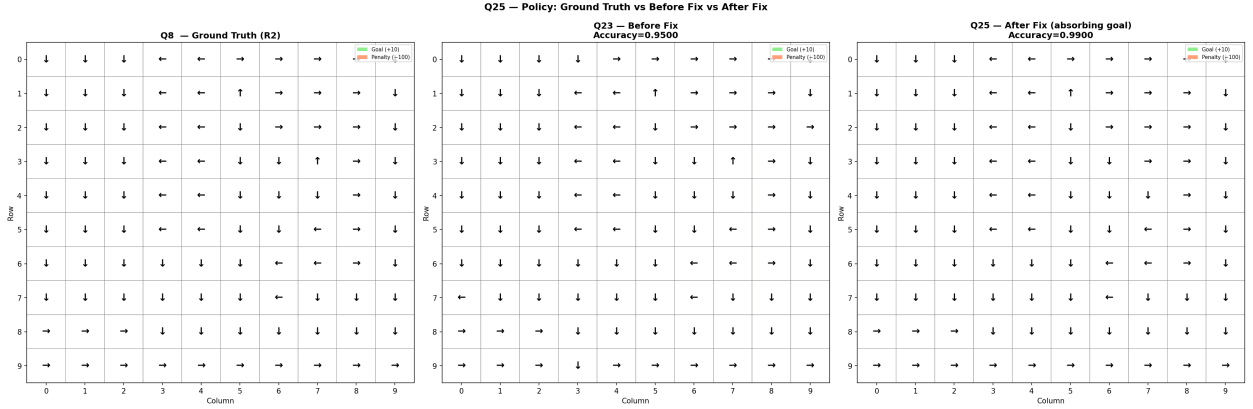


Figure 26: Three-way policy comparison: Q8 ground truth (left), Q23 before fix with accuracy 0.95 (center), and Q25 after fix with accuracy 0.99 (right).

	Max accuracy	λ at maximum
Before fix	0.9500	1.3627
After fix	0.9900	0.5511

Table 1: Accuracy improvement after applying the absorbing goal fix.

Result. The absorbing goal fix improves the maximum accuracy from 0.95 to 0.99 (+0.04), confirming that Discrepancy 1 was caused by the non-absorbing goal state in the value iteration. The remaining 1% mismatch corresponds to Discrepancy 2, which persists because it originates from

the reward scale ambiguity inherent to the LP-based IRL formulation and cannot be resolved by modifying the value iteration alone.